

SOFTWARE PROJECT DESCRIPTION (Summarized)

PROJECT NAME

Smart Hospital Management System

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PROJECT OVERVIEW

The Smart Hospital Management System is a database-based system designed to support and improve the daily operations of a hospital. The system helps manage important hospital information such as patients, staff, appointments, medical records, treatments, pharmacy services, laboratory services, billing, and payments. Instead of storing records manually, the system allows hospital staff to store, update, search, and manage information more efficiently and securely. The system consists of six main modules: Core System Management, Appointment Management, Medical Record and Treatment Management, Pharmacy Management, Laboratory and Diagnostic Management, and Billing and Payment Management. These modules work together to support the hospital workflow from patient registration to treatment, medicine dispensing, lab testing, and payment. This system is useful because it reduces paperwork, minimizes data duplication, improves record accuracy, and helps hospital staff access information faster. It also supports better communication between departments such as reception, doctors, pharmacy, laboratory, and billing. Overall, the Smart Hospital Management System provides a structured and reliable solution for managing hospital services in a more modern and efficient way.

PROJECTS

Module #01 - Core System

Patient Management

Function #1 - Search	Search patients by name, ID, or contact info
Function #2 - Register new patient	Create a new patient record with personal info, insurance info
Function #3 - View patient profile	View full patient details, history, and assigned doctor
Function #4 - Update patient info	Edit existing patient personal information, For example: contact, address..
Function #5 - Discharge patient	Process patient discharge and update status
Function #6 - View Medical history	View past visits, diagnoses, prescriptions, and lab results
Function #7 - Assign patient to room	Link a patient to a specific room or bed

Staff Management - Doctor

Function #1 - Search	Search doctor by name, ID, or specialisation
Function #2 - Add doctor profile	Register a new doctor with personal and professional info

Function #3 - Update doctor info	Edit doctor details, specialisation, or contact
Function #4 - View doctor schedule	View assigned appointments and shifts for a doctor
Function #5 - Assign to department	Link doctor to a specific department
Function #6 - Deactivate account	Disable a doctor account without deleting records

Staff Management - Nurse

Function #1 - Search	Search nurse by name, ID, or ward/bed
Function #2 - Add nurse profile	Register a new nurse with personal and professional info
Function #3 - Update nurse info	Edit nurse details or contact information
Function #4 - Assign to ward	Assign a nurse to a specific ward or department
Function #5 - View shift schedule	View nurse working shifts and assignments
Function #6 - Deactivate account	Disable a nurse account without deleting records

Staff Management - Receptionist

Function #1 - Search	Search nurse by name, ID
Function #2 - Add receptionist profile	Register a new receptionist with personal info
Function #3 - Update receptionist info	Edit receptionist details or contact information
Function #4 - Assign to counter	Assign a receptionist to a specific counter or desk
Function #5 - View schedule	View receptionist working schedule
Function #6 - Deactivate account	Disable a receptionist account without deleting records

Staff Management - Pharmacist

Function #1 - Search	Search pharmacist by name, ID
Function #2 - Add pharmacist profile	Register a new pharmacist with personal and license info
Function #3 - Update pharmacist info	Edit pharmacist details or contact information
Function #4 - Assign to counter	Assign a pharmacist to a specific pharmacist unit
Function #5 - View schedule	View pharmacist working schedule

Function #6 - Deactivate account	Disable a pharmacist account without deleting records
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Staff Management - Lab Technician

Function #1 - Search	Search lab technician by name, ID
Function #2 - Add lab technician profile	Register a new lab technician with personal info
Function #3 - Update lab technician info	Edit lab technician details or contact information
Function #4 - Assign to lab	Assign a lab technician to a specific laboratory
Function #5 - View schedule	View lab technician working schedule
Function #6 - Deactivate account	Disable a lab technician account without deleting records

Department Management

Function #1 - Search	Search department by name or type
Function #2 - Create department	Add a new department with name and description
Function #3 - Update department	Edit department name, head, or details

Function #4 - Assign Staff	Link staff members to a department
Function #5 - Transfer patient	Move a patient from one department to another
Function #6 - View occupancy	Check current patient count and capacity per department
Function #7 - Generate report	Export department statistics and performance report

Room Management

Function #1 - Search	Search rooms by type, floor, or availability status
Function #2 - Add room	Register a new room with type, floor, and capacity
Function #3 - Update info	Edit room details or type classification
Function #4 - Assign patient	Link a patient to a specific room or bed or ward
Function #5 - Check availability	View real-time room and bed availability
Function #6 - Mark for maintenance	Flag a room for cleaning or maintenance work
Function #7 - Release room	Mark room as available after patient discharge

Schedule Management

Function #1 - Search	Search appointments by patient, doctor, or date
Function #2 - Create appointment slot	Define available time slots for doctors
Function #3 - Book appointment	Schedule available time slots for doctors
Function #4 - Cancel appointment	Cancel an existing appointment with reason log
Function #5 - Reschedule Appointment	Move an appointment to a different date or time
Function #6 - Manage staff shifts	Assign and update staff working schedules
Function #7 - View calendar (nice to have)	View daily or weekly schedule for all staff
Function #8 - Send reminder	Send appointment reminder to patient or doctor

Module #02 - Appointment Management

Function #1 - Search appointments by patient, doctor, or date	Allow staff to retrieve appointment records using search criteria.
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Function #2 - Book Appointment	Allow staff to schedule appointments for patients.
Function #3 - Update Appointment	Allow staff to modify appointment details.
Function #4 - Cancel Appointment	Allow staff to cancel the appointment before completion
Function #5 - Check Doctor Availability	Allow staff to check available doctor schedules before booking.
Function #6 - View Appointment	Allow staff to view appointment details and schedule

Module #03 - Medical Record and Treatment Management

Medical Record Management

Function #1 - Search Medical Records	Retrieve patient medical records using patient ID, patient name, doctor name, or visit date.
Function #2 - Create Medical Record	Create a new medical record for a patient during consultation, including symptoms, diagnosis, and treatment notes.
Function #3 - View Medical History	Display a patient's previous visits, diagnoses, prescriptions, treatments, allergies, and laboratory results.
Function #4 - Adjust Medical Record	Allow authorized medical staff to make corrections or add amendments to medical records while recording: • adjustment reason • adjusted by (staff ID) • adjustment date and time The old version still keep in database

Function #5 - Record Vital Signs	Store patient vital signs such as temperature, blood pressure, heart rate, height, and weight.
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Treatment Management

Function #1 -Record Diagnosis & Treatment Plan	Record the doctor's diagnosis and create a structured treatment plan for the patient, including clinical notes and recommended care.
Function #2 – Manage Prescription	allows doctors to create and manage prescriptions, including medicine name, dosage, frequency, duration, and usage instructions.
Function #3 - Record Medical Procedures	documents medical procedures or treatments performed on the patient, including procedure details and outcomes.
Function #4 - Assign Doctor to Case	Link a doctor to a patient's consultation or treatment record.
Function #5 - Upload Laboratory Results	Attach laboratory test results or diagnostic reports to the patient's medical record.
Function #6 – Generate Medical Report	Generate and export patient medical summaries, treatment reports, or consultation history.

Module #04 - Pharmacy and Medicine Inventory

Function #1 - Medicine Inventory Tracking System	Tracking and managing the stocks of the medicine.
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Function #2 - Medication Dispensing	Issues the medicine to patients, updates the inventory, and generates a dispensing record.
Function #3 - Batch Expiry Date Tracking	Track medicine batches and expiration dates and alert if the expiration date is near.
Function #4 - Drug Interaction and Allergy Checking	Check for possible drug interaction and possible patient allergies.
Function #5 - Drug Substitution	If a medicine is out of stock, suggest alternative medicines.

Module #05 - Laboratory and Diagnostic Management

Function #1 - Test Order Management	Assign Task to Lab Technician, and Track test order status.
Function #2 - Test Result Entry	Input Lab results into the system.
Function #3 - Equipment & Resource Tracking	Track lab equipment usage and available lab equipment.
Function #4 - Historical Result Comparison	Compare current result with past records
Function #5 - Generate Lab report	Generates and exports a complete laboratory report containing test results and diagnostic information for doctors and patients.

Module #06 - Billing and Payment

Function #1 - Generate Bill1	Create a bill after appointment/treatment.
Function #2 - Add Bill Items	Add services, medicines, lab tests, or other charges to the bill.
Function #3 - View Bill	Allow staff to view bill details and total charges.
Function #4 - Process Payment	Allow staff to record patient payments.
Function #5 - View Payment History	Allow staff to review previous payment transactions.

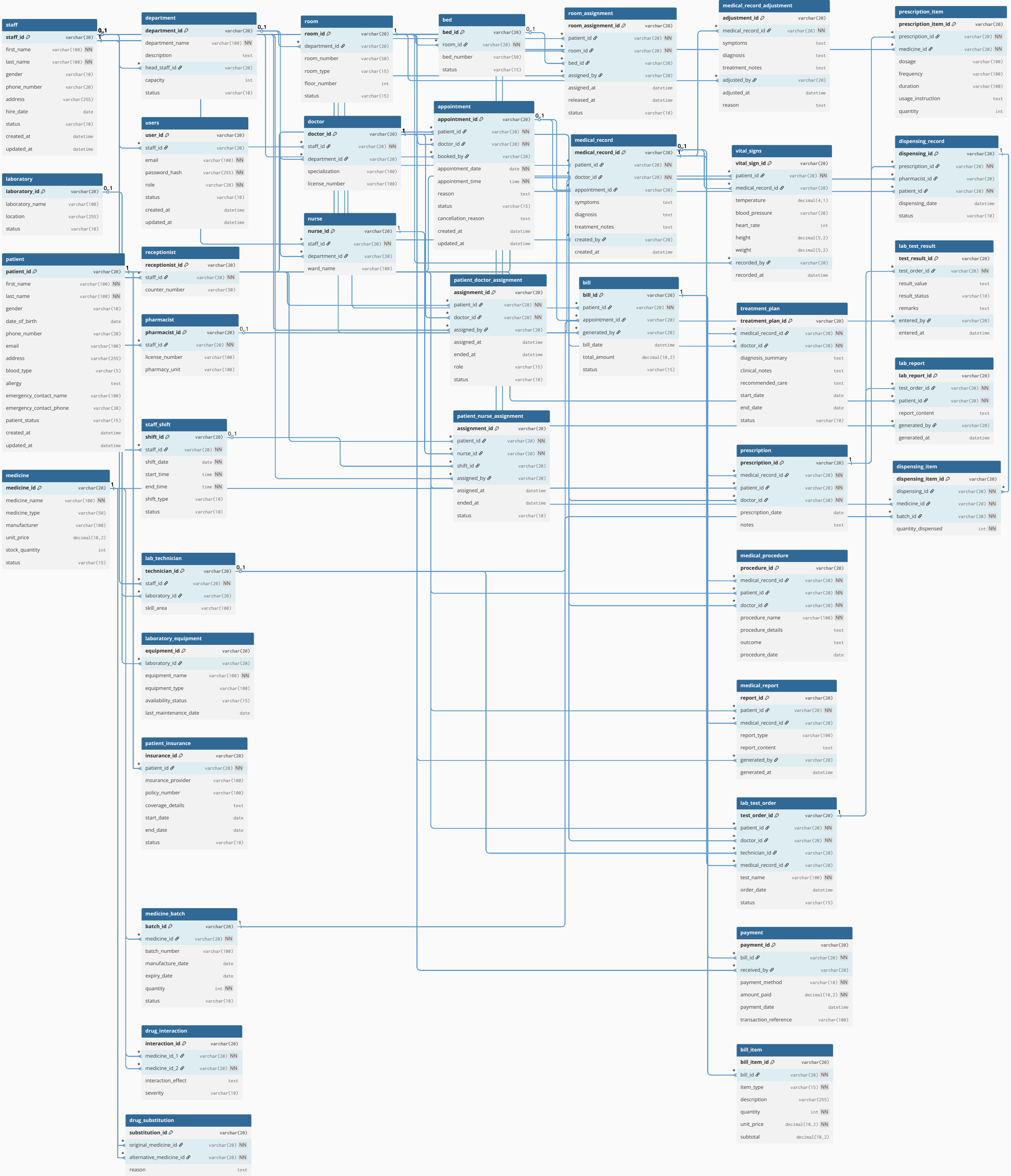


TABLE DEFINITION

1. Users

Stores login accounts for system users.

Column	Type	Key	Description
user_id	VARCHAR(20)	PK	Unique user ID
staff_id	VARCHAR(20)	FK	Linked staff
email	VARCHAR(100)	UNIQUE	Login email
password_hash	VARCHAR(255)		Encrypted password
role	ENUM		User role, 'admin','doctor','nurse','receptionist','pharmacist','lab_technician'
status	ENUM		Account status: 'active','inactive'
created_at	DATETIME		Account created date and time
updated_at	DATETIME		Last updated date and time

2. Staff

Stores common staff personal information.

Column	Type	Key	Description
staff_id	VARCHAR(20)	PK	Unique staff ID
first_name	VARCHAR(100)		Staff first name
last_name	VARCHAR(100)		Staff last name
gender	ENUM		Staff gender ('male','female','other')
phone_number	VARCHAR(20)		Contact number

address	VARCHAR(255)		Staff address
hire_date	DATE		Date staff joined
status	ENUM		Staff status ('active','inactive')
created_at	DATETIME		Created date and time
updated_at	DATETIME		Last updated date and time

3. Department

Stores hospital departments.

Column	Type	Key	Description
department_id	VARCHAR(20)	PK	Unique department ID
department_name	VARCHAR(100)		Department name
description	TEXT		Department description
head_staff_id	VARCHAR(20)	FK	Department head
capacity	INT		Maximum patient capacity
status	ENUM		Department status ('active','inactive')

4. Doctor

Stores doctor-specific information.

Column	Type	Key	Description
doctor_id	VARCHAR(20)	PK	Unique doctor ID
staff_id	VARCHAR(20)	FK, UNIQUE	Linked staff
department_id	VARCHAR(20)	FK	Assigned department
specialization	VARCHAR(100)		Doctor specialization
license_number	VARCHAR(100)	UNIQUE	Medical license number

5. Nurse

Stores nurse-specific information.

Column	Type	Key	Description
nurse_id	VARCHAR(20)	PK	Unique nurse ID
staff_id	VARCHAR(20)	FK, UNIQUE	Linked staff
department_id	VARCHAR(20)	FK	Assigned department
ward_name	VARCHAR(100)		Assigned ward

6. Receptionist

Stores receptionist-specific information.

Column	Type	Key	Description
receptionist_id	VARCHAR(20)	PK	Unique receptionist ID
staff_id	VARCHAR(20)	FK, UNIQUE	Linked staff
counter_number	VARCHAR(50)		Assigned counter

7. Pharmacist

Stores pharmacist-specific information.

Column	Type	Key	Description
pharmacist_id	VARCHAR(20)	PK	Unique pharmacist ID
staff_id	VARCHAR(20)	FK, UNIQUE	Linked staff
license_number	VARCHAR(100)	UNIQUE	Pharmacist license number
pharmacy_unit	VARCHAR(100)		Assigned pharmacy unit

8. Laboratory

Stores laboratory information.

Column	Type	Key	Description
laboratory_id	VARCHAR(20)	PK	Unique laboratory ID
laboratory_name	VARCHAR(100)		Laboratory name
location	VARCHAR(255)		Laboratory location
status	ENUM('active','inactive')		Laboratory status

9. Lab_Technician

Stores lab technician-specific information.

Column	Type	Key	Description
technician_id	VARCHAR(20)	PK	Unique technician ID
staff_id	VARCHAR(20)	FK, UNIQUE	Linked staff
laboratory_id	VARCHAR(20)	FK	Assigned laboratory
skill_area	VARCHAR(100)		Lab technician skill area

10. Patient

Stores patient information.

Column	Type	Key	Description
patient_id	VARCHAR(20)	PK	Unique patient ID
first_name	VARCHAR(100)		Patient first name
last_name	VARCHAR(100)		Patient last name
gender	ENUM		Patient gender ('male','female','other')
date_of_birth	DATE		Patient date of birth
phone_number	VARCHAR(20)		Contact number
email	VARCHAR(100)		Email address
address	VARCHAR(255)		Patient address
blood_type	VARCHAR(5)		Blood type
allergy	TEXT		Patient allergy information
emergency_contact_name	VARCHAR(100)		Emergency contact name
emergency_contact_phone	VARCHAR(20)		Emergency contact phone
patient_status	ENUM		Patient status ('active','admitted','discharged')
created_at	DATETIME		Registration date and time
updated_at	DATETIME		Last updated date and time

11. Patient_Insurance

Stores patient insurance information.

Column	Type	Key	Description
insurance_id	VARCHAR(20)	PK	Unique insurance ID
patient_id	VARCHAR(20)	FK	Linked patient
insurance_provider	VARCHAR(100)		Insurance provider name
policy_number	VARCHAR(100)		Insurance policy number
coverage_details	TEXT		Coverage information
start_date	DATE		Insurance start date
end_date	DATE		Insurance end date
status	ENUM		Insurance status ('active','expired','cancelled')

12. Room

Stores hospital rooms.

Column	Type	Key	Description
room_id	VARCHAR(20)	PK	Unique room ID
department_id	VARCHAR(20)	FK	Department location
room_number	VARCHAR(50)		Room number
room_type	ENUM		Room type ('general','private','icu','emergency')
floor_number	INT		Floor number
status	ENUM		Room status ('available','occupied','maintenance')

13. Bed

Stores beds inside rooms.

Column	Type	Key	Description
bed_id	VARCHAR(20)	PK	Unique bed ID
room_id	VARCHAR(20)	FK	Linked room
bed_number	VARCHAR(50)		Bed number
status	ENUM		Bed status ('available','occupied','maintenance')

14. Room_Assignment

Stores patient room and bed assignment history.

Column	Type	Key	Description
room_assignment_id	VARCHAR(20)	PK	Unique room assignment ID
patient_id	VARCHAR(20)	FK	Linked patient
room_id	VARCHAR(20)	FK	Assigned room
bed_id	VARCHAR(20)	FK	Assigned bed
assigned_by	VARCHAR(20)	FK	Staff who assigned room
assigned_at	DATETIME		Assignment date and time
released_at	DATETIME		Release date and time
status	ENUM		Assignment status ('active','completed','cancelled')

15. Appointment

Stores patient appointments.

Column	Type	Key	Description
appointment_id	VARCHAR(20)	PK	Unique appointment ID
patient_id	VARCHAR(20)	FK	Linked patient
doctor_id	VARCHAR(20)	FK	Assigned doctor
booked_by	VARCHAR(20)	FK	Staff who booked appointment
appointment_date	DATE		Appointment date
appointment_time	TIME		Appointment time
reason	TEXT		Reason for visit
status	ENUM		Appointment status ('scheduled','completed','cancelled')
cancellation_reason	TEXT		Reason if cancelled
created_at	DATETIME		Booking date and time
updated_at	DATETIME		Last updated date and time

16. Staff_Shift

Stores staff working schedule.

Column	Type	Key	Description
shift_id	VARCHAR(20)	PK	Unique shift ID
staff_id	VARCHAR(20)	FK	Linked staff
shift_date	DATE		Working date
start_time	TIME		Shift start time
end_time	TIME		Shift end time
shift_type	ENUM		Shift type 'morning','afternoon','night')
status	ENUM		Shift status ('scheduled','completed','cancelled')

17. Patient_Doctor_Assignment

Links patients with doctors.

Column	Type	Key	Description
assignment_id	VARCHAR(20)	PK	Unique assignment ID
patient_id	VARCHAR(20)	FK	Linked patient
doctor_id	VARCHAR(20)	FK	Assigned doctor
assigned_by	VARCHAR(20)	FK	Staff who assigned doctor
assigned_at	DATETIME		Assignment date and time
ended_at	DATETIME		Assignment ended date and time
role	ENUM		Doctor role for patient ('main_doctor','consultant','specialist')

status	ENUM		Assignment status ('active','completed','cancelled')
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18. Patient_Nurse_Assignment

Links patients with nurses.

Column	Type	Key	Description
assignment_id	VARCHAR(20)	PK	Unique assignment ID
patient_id	VARCHAR(20)	FK	Linked patient
nurse_id	VARCHAR(20)	FK	Assigned nurse
shift_id	VARCHAR(20)	FK	Related shift
assigned_by	VARCHAR(20)	FK	Staff who assigned nurse
assigned_at	DATETIME		Assignment date and time
ended_at	DATETIME		Assignment ended date and time
status	ENUM		Assignment status ('active','completed','cancelled')

19. Medical_Record

Stores original medical records.

Column	Type	Key	Description
medical_record_id	VARCHAR(20)	PK	Unique medical record ID
patient_id	VARCHAR(20)	FK	Linked patient
doctor_id	VARCHAR(20)	FK	Responsible doctor
appointment_id	VARCHAR(20)	FK	Related appointment

symptoms	TEXT		Patient symptoms
diagnosis	TEXT		Diagnosis
treatment_notes	TEXT		Treatment notes
created_by	VARCHAR(20)	FK	Staff who created record
created_at	DATETIME		Record creation date and time

20. Medical_Record_Adjustment

Stores edited versions of medical records.

Column	Type	Key	Description
adjustment_id	VARCHAR(20)	PK	Unique adjustment ID
medical_record_id	VARCHAR(20)	FK	Linked original medical record
symptoms	TEXT		Updated symptoms
diagnosis	TEXT		Updated diagnosis
treatment_notes	TEXT		Updated treatment notes
adjusted_by	VARCHAR(20)	FK	Staff who adjusted record
adjusted_at	DATETIME		Adjustment date and time
reason	TEXT		Reason for adjustment

21. Vital_Signs

Stores patient vital signs.

Column	Type	Key	Description
vital_sign_id	VARCHAR(20)	PK	Unique vital sign ID
patient_id	VARCHAR(20)	FK	Linked patient
medical_record_id	VARCHAR(20)	FK	Related medical record

temperature	DECIMAL(4,1)		Body temperature
blood_pressure	VARCHAR(20)		Blood pressure
heart_rate	INT		Heart rate
height	DECIMAL(5,2)		Patient height
weight	DECIMAL(5,2)		Patient weight
recorded_by	VARCHAR(20)	FK	Staff who recorded
recorded_at	DATETIME		Record date and time

22. Treatment_Plan

Stores diagnosis and treatment plan.

Column	Type	Key	Description
treatment_plan_id	VARCHAR(20)	PK	Unique treatment plan ID
medical_record_id	VARCHAR(20)	FK	Linked medical record
doctor_id	VARCHAR(20)	FK	Doctor who created plan
diagnosis_summary	TEXT		Diagnosis summary
clinical_notes	TEXT		Clinical notes
recommended_care	TEXT		Recommended care
start_date	DATE		Treatment start date
end_date	DATE		Treatment end date
status	ENUM		Treatment status ('active','completed','cancelled')

23. Prescription

Stores prescription header.

Column	Type	Key	Description
prescription_id	VARCHAR(20)	PK	Unique prescription ID
medical_record_id	VARCHAR(20)	FK	Related medical record
patient_id	VARCHAR(20)	FK	Linked patient
doctor_id	VARCHAR(20)	FK	Prescribing doctor
prescription_date	DATE		Prescription date
notes	TEXT		Additional notes

24. Prescription_Item

Stores medicine items in prescription.

Column	Type	Key	Description
prescription_item_id	VARCHAR(20)	PK	Unique prescription item ID
prescription_id	VARCHAR(20)	FK	Linked prescription
medicine_id	VARCHAR(20)	FK	Linked medicine
dosage	VARCHAR(100)		Dosage
frequency	VARCHAR(100)		Usage frequency
duration	VARCHAR(100)		Duration
usage_instruction	TEXT		Usage instructions
quantity	INT		Quantity prescribed

25. Medical_Procedure

Stores medical procedures performed.

Column	Type	Key	Description
procedure_id	VARCHAR(20)	PK	Unique procedure ID
medical_record_id	VARCHAR(20)	FK	Related medical record
patient_id	VARCHAR(20)	FK	Linked patient
doctor_id	VARCHAR(20)	FK	Doctor responsible
procedure_name	VARCHAR(100)		Procedure name
procedure_details	TEXT		Procedure details
outcome	TEXT		Procedure outcome
procedure_date	DATE		Procedure date

26. Medical_Report

Stores generated medical reports.

Column	Type	Key	Description
report_id	VARCHAR(20)	PK	Unique report ID
patient_id	VARCHAR(20)	FK	Linked patient
medical_record_id	VARCHAR(20)	FK	Related medical record
report_type	VARCHAR(100)		Summary, treatment, or history
report_content	TEXT		Report details
generated_by	VARCHAR(20)	FK	Staff who generated report
generated_at	DATETIME		Report generation date and time

27. Medicine

Stores medicine information.

Column	Type	Key	Description
medicine_id	VARCHAR(20)	PK	Unique medicine ID
medicine_name	VARCHAR(100)		Medicine name
medicine_type	VARCHAR(50)		Tablet, syrup, injection
manufacturer	VARCHAR(100)		Manufacturer name
unit_price	DECIMAL(10,2)		Price per unit
stock_quantity	INT		Current total stock
status	ENUM		Medicine status ('available','unavailable')

28. Medicine_Batch

Stores medicine batch and expiry information.

Column	Type	Key	Description
batch_id	VARCHAR(20)	PK	Unique batch ID
medicine_id	VARCHAR(20)	FK	Linked medicine
batch_number	VARCHAR(100)		Batch number
manufacture_date	DATE		Manufacture date
expiry_date	DATE		Expiry date
quantity	INT		Quantity in batch
status	ENUM		Batch status

29. Dispensing_Record

Stores medicine dispensing records.

Column	Type	Key	Description
dispensing_id	VARCHAR(20)	PK	Unique dispensing ID
prescription_id	VARCHAR(20)	FK	Linked prescription
pharmacist_id	VARCHAR(20)	FK	Pharmacist who dispensed
patient_id	VARCHAR(20)	FK	Linked patient
dispensing_date	DATETIME		Dispensing date and time
status	ENUM		Dispensing status ('dispensed','cancelled')

30. Dispensing_Item

Stores dispensed medicine details.

Column	Type	Key	Description
dispensing_item_id	VARCHAR(20)	PK	Unique dispensing item ID
dispensing_id	VARCHAR(20)	FK	Linked dispensing record
medicine_id	VARCHAR(20)	FK	Linked medicine
batch_id	VARCHAR(20)	FK	Linked medicine batch
quantity_dispensed	INT		Dispensed quantity

31. Drug_Interaction

Stores medicine interaction rules.

Column	Type	Key	Description
interaction_id	VARCHAR(20)	PK	Unique interaction ID
medicine_id_1	VARCHAR(20)	FK	First medicine
medicine_id_2	VARCHAR(20)	FK	Second medicine
interaction_effect	TEXT		Interaction effect
severity	ENUM		Interaction severity ('low','medium','high')

32. Drug_Substitution

Stores alternative medicine options.

Column	Type	Key	Description
substitution_id	VARCHAR(20)	PK	Unique substitution ID
original_medicine_id	VARCHAR(20)	FK	Original medicine
alternative_medicine_id	VARCHAR(20)	FK	Alternative medicine
reason	TEXT		Reason for substitution

33. Lab_Test_Order

Stores laboratory test orders.

Column	Type	Key	Description
test_order_id	VARCHAR(20)	PK	Unique test order ID
patient_id	VARCHAR(20)	FK	Linked patient

doctor_id	VARCHAR(20)	FK	Doctor who ordered test
technician_id	VARCHAR(20)	FK	Assigned lab technician
medical_record_id	VARCHAR(20)	FK	Related medical record
test_name	VARCHAR(100)		Test name
order_date	DATETIME		Order date and time
status	ENUM		Test order status ('pending','in_progress','completed','cancelled')

34. Lab_Test_Result

Stores laboratory test results.

Column	Type	Key	Description
test_result_id	VARCHAR(20)	PK	Unique test result ID
test_order_id	VARCHAR(20)	FK	Linked test order
result_value	TEXT		Test result value/details
result_status	ENUM		Result status ('normal','abnormal')
remarks	TEXT		Additional remarks
entered_by	VARCHAR(20)	FK	Technician who entered result
entered_at	DATETIME		Result entry date and time

35. Laboratory_Equipment

Stores laboratory equipment.

Column	Type	Key	Description
equipment_id	VARCHAR(20)	PK	Unique equipment ID
laboratory_id	VARCHAR(20)	FK	Linked laboratory
equipment_name	VARCHAR(100)		Equipment name
equipment_type	VARCHAR(100)		Equipment type
availability_status	ENUM		Equipment availability('available','in_use','maintenance')
last_maintenance_date	DATE		Last maintenance date

36. Lab_Report

Stores generated lab reports.

Column	Type	Key	Description
lab_report_id	VARCHAR(20)	PK	Unique lab report ID
test_order_id	VARCHAR(20)	FK	Linked test order
patient_id	VARCHAR(20)	FK	Linked patient
report_content	TEXT		Lab report details
generated_by	VARCHAR(20)	FK	Staff who generated report
generated_at	DATETIME		Report generation date and time

37. Bill

Stores patient bill.

Column	Type	Key	Description
bill_id	VARCHAR(20)	PK	Unique bill ID
patient_id	VARCHAR(20)	FK	Linked patient
appointment_id	VARCHAR(20)	FK	Related appointment
generated_by	VARCHAR(20)	FK	Staff who generated bill
bill_date	DATETIME		Bill creation date and time
total_amount	DECIMAL(10,2)		Total bill amount
status	ENUM		Bill status ('unpaid','partially_paid','paid')

38. Bill_Item

Stores bill details.

Column	Type	Key	Description
bill_item_id	VARCHAR(20)	PK	Unique bill item ID
bill_id	VARCHAR(20)	FK	Linked bill
item_type	ENUM		Type of billed item ('service','medicine','lab_test','procedure','room')
description	VARCHAR(255)		Item description
quantity	INT		Quantity
unit_price	DECIMAL(10,2)		Unit price
subtotal	DECIMAL(10,2)		Quantity × unit price

39. Payment

Stores payment transactions.

Column	Type	Key	Description
payment_id	VARCHAR(20)	PK	Unique payment ID
bill_id	VARCHAR(20)	FK	Linked bill
received_by	VARCHAR(20)	FK	Staff who received payment
payment_method	ENUM		Payment method ('cash','card','online')
amount_paid	DECIMAL(10,2)		Payment amount
payment_date	DATETIME		Payment date and time
transaction_reference	VARCHAR(100)		Transaction/reference number